

DMU Recap

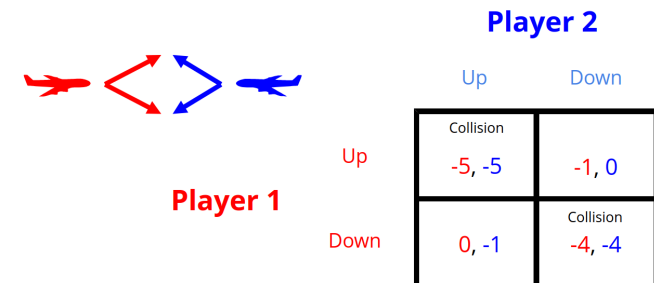
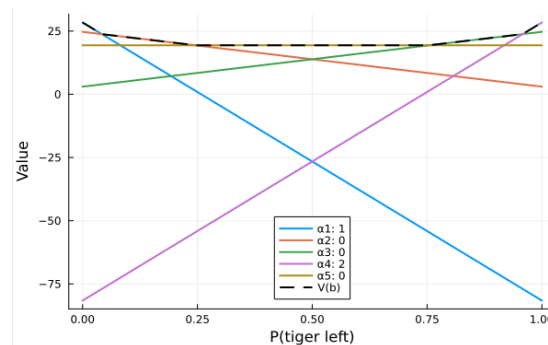
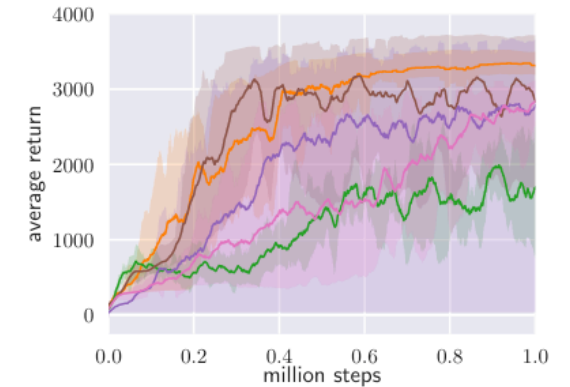
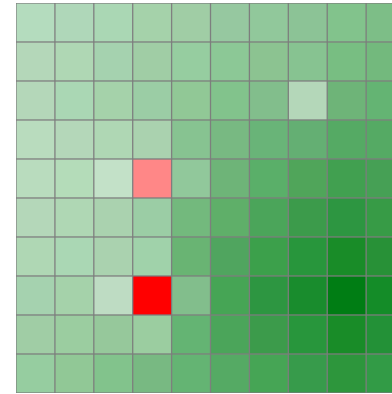
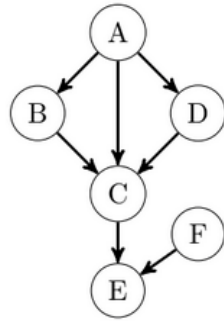
Probabilistic
Models

Markov Decision
Processes

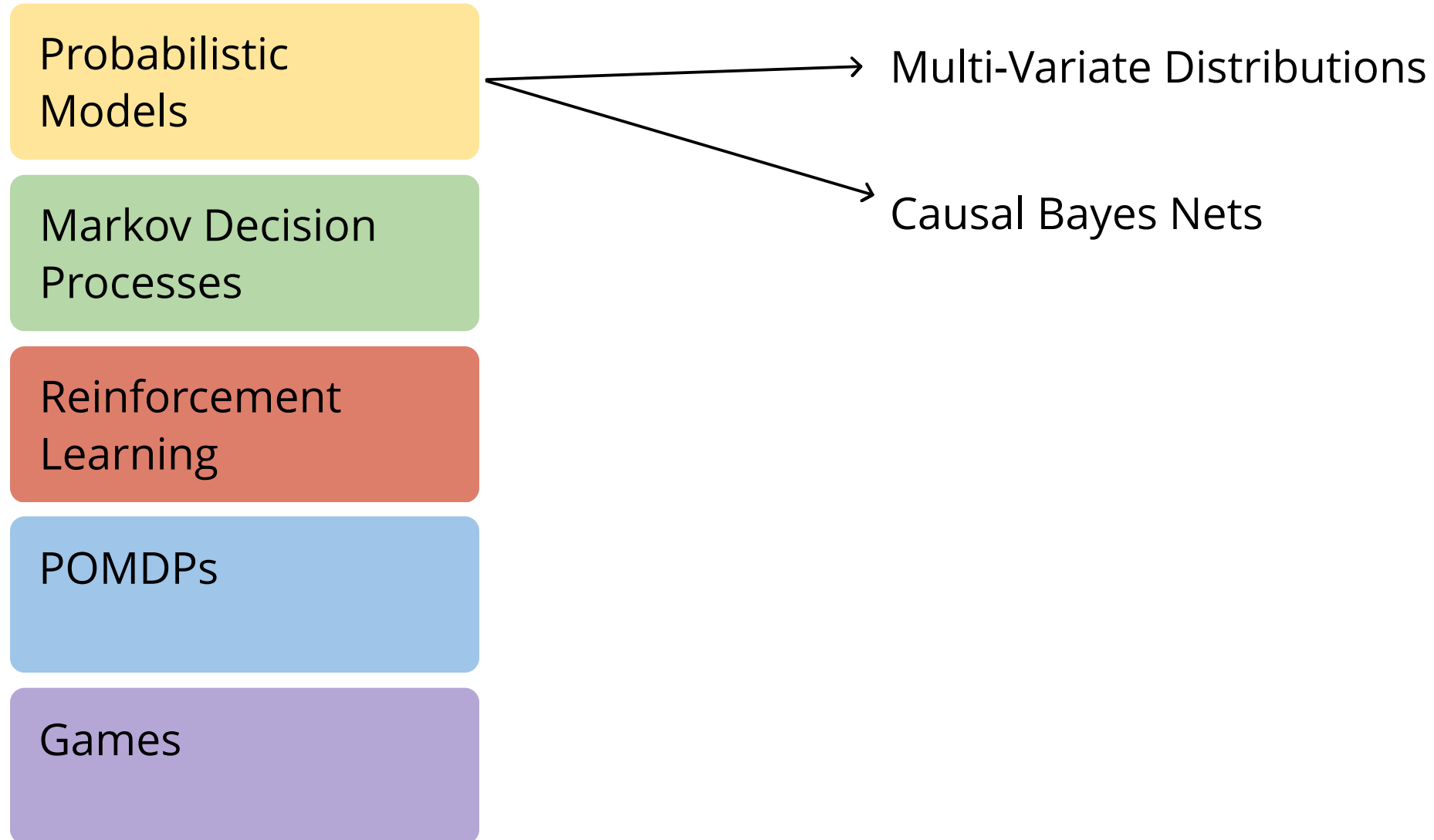
Reinforcement
Learning

POMDPs

Games



DMU Recap



Multi-Variate Distributions

Marginal: $P(A)$

Joint: $P(A, B)$

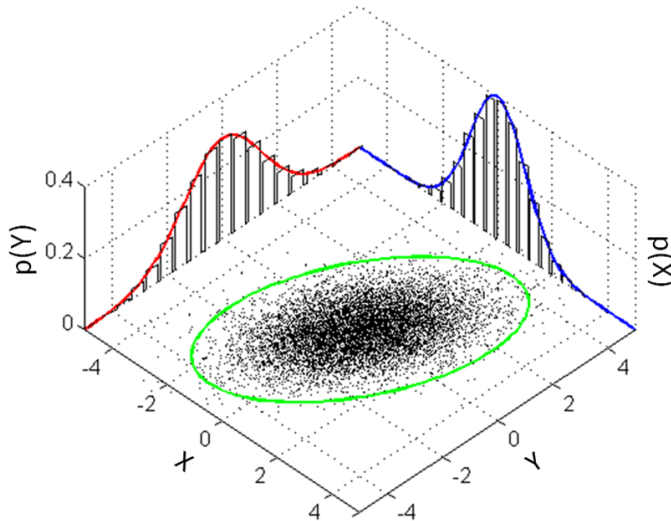
Conditional: $P(A | B)$

3 Rules

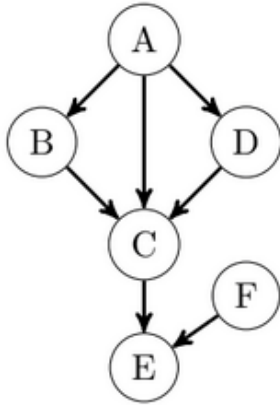
$$1. 0 \leq P(X | Y) \leq 1$$
$$\sum_{x \in X} P(x | Y) = 1$$

$$2. P(X) = \sum_{y \in Y} P(X, y)$$

$$3. P(X | Y) = \frac{P(X, Y)}{P(Y)}$$



Causal Bayesian Networks



Each node is a **random variable** with a defined **distribution conditioned** on parents

Chain Rule

$$P(X_{1:n}) = \prod_i P(X_i \mid Pa(X_i))$$

(Naive) Inference

$$P(Q = q \mid E = e) = \frac{P(Q=q, E=e)}{P(E=e)} = \frac{\sum_i P(Q=q, I=i, E=e)}{\sum_{i,q} P(Q=q, I=i, E=e)}$$

Apply chain rule based on structure

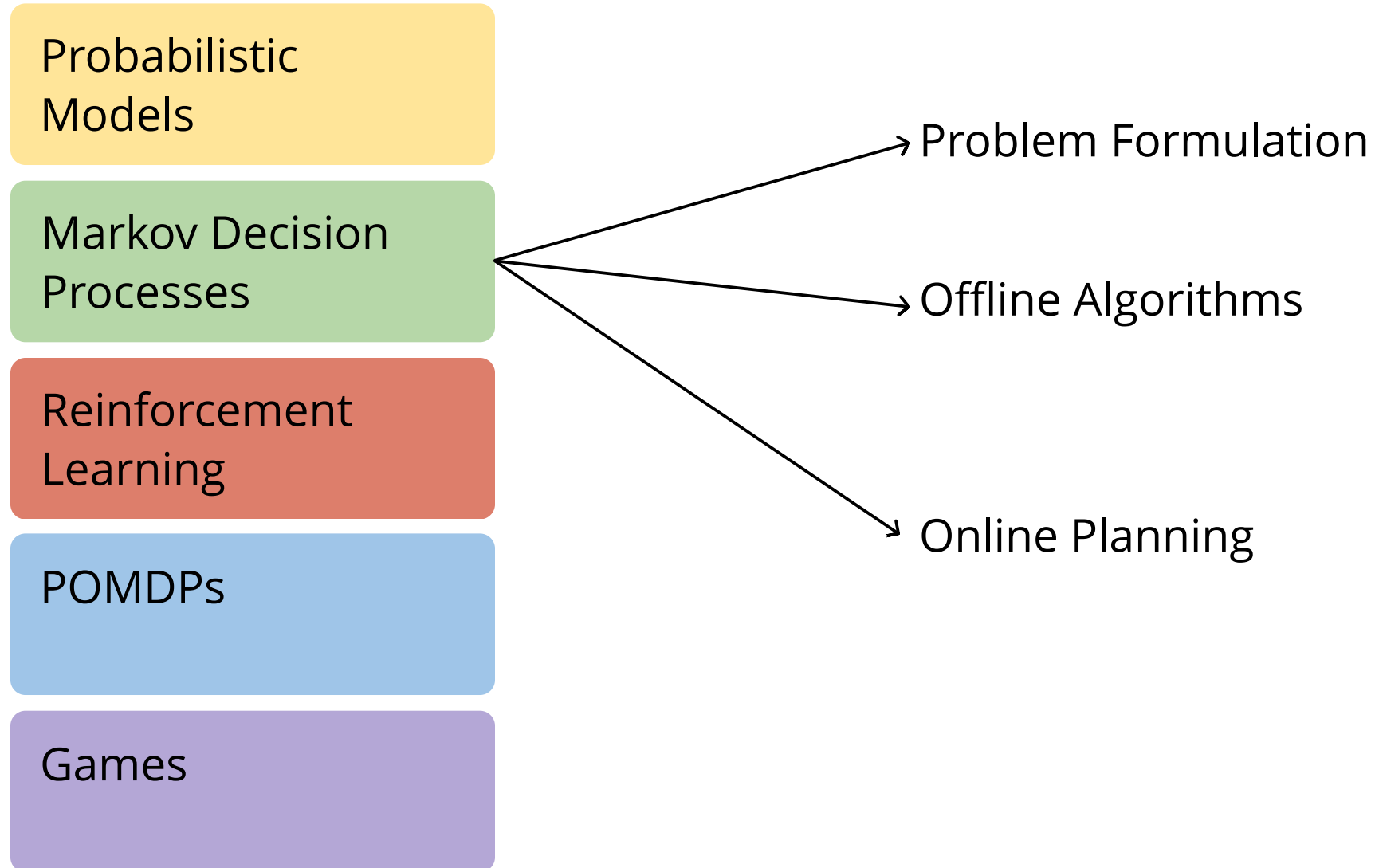
Special Case: Bayes Rule

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

Conditional Independence

$X \perp Y \mid \mathcal{C}$ if all paths between X and Y are d-separated by $\mathcal{C}$

DMU Recap



Markov Decision Processes

Examples: $S = \{1, 2, 3\}$ or $S = \mathbb{R}^2$

$$s = (x, \dot{x}) \in S = \mathbb{R}^2$$

$$Q^\pi(s, a) = E \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s, a_0 = a, a_t = \pi(s_t) \right]$$

$$V^\pi(s) = Q^\pi(s, \pi(s))$$

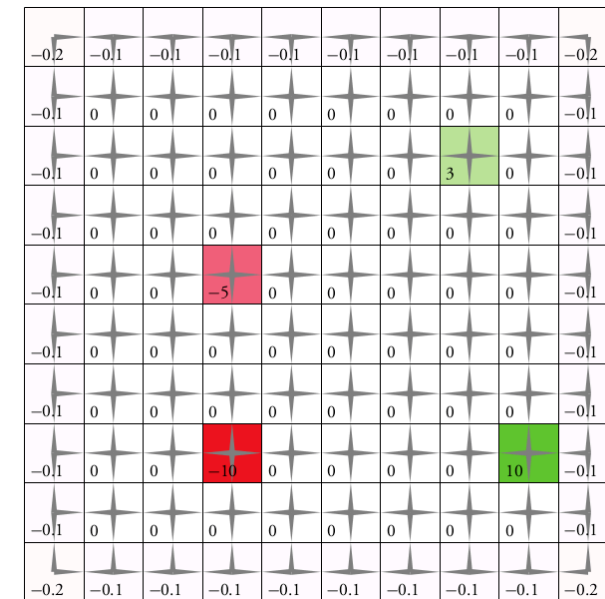
$$V^\pi(s) = R(s, a) + \gamma E[V^\pi(s')]$$

$$V^*(s) = \max_a \{R(s, a) + \gamma E[V^*(s')]\}$$

$$B[V](s) = \max_a \{R(s, a) + \gamma E[V(s')]\}$$

$$(S, A, R, T, \gamma)$$

$$\underset{\pi}{\text{maximize}} E \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right]$$

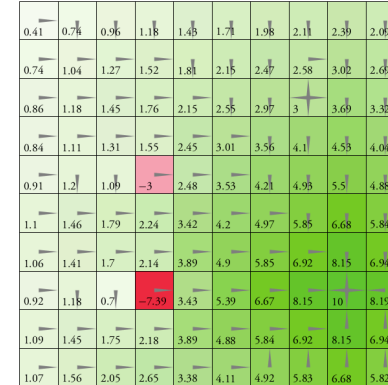
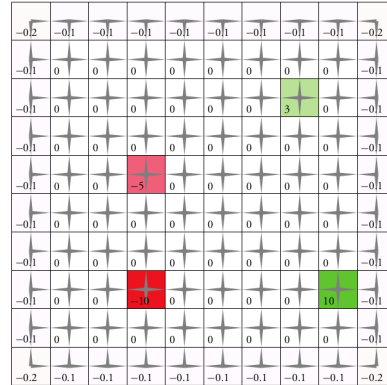


Policy Evaluation

Bellman's Equation: Certificate of Optimality

Bellman's Operator

Offline MDP Algorithms



Policy Iteration

loop

Evaluate Policy

Improve Policy

Finds optimal policy because
policy always improves and there
are a finite number of policies

Value Iteration

loop

$$V \leftarrow B[V]$$

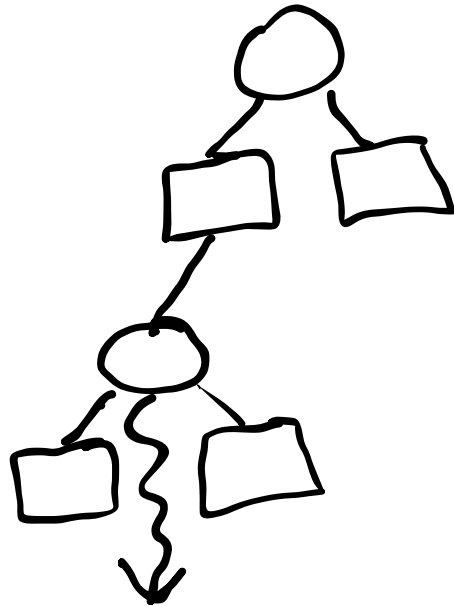
Converges because B is
a contraction mapping

Online MDP Planning

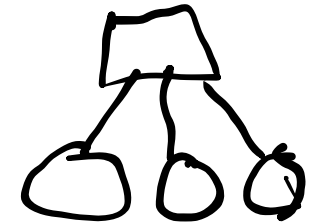
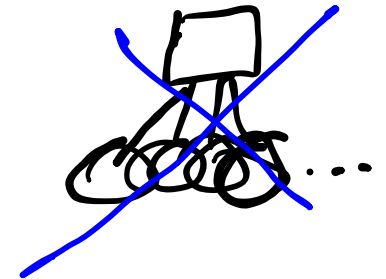
Monte Carlo Tree Search

Search
Expand
Rollout
Backup

$$Q(s,a) + c \sqrt{\frac{\log N(s)}{N(s,a)}}$$

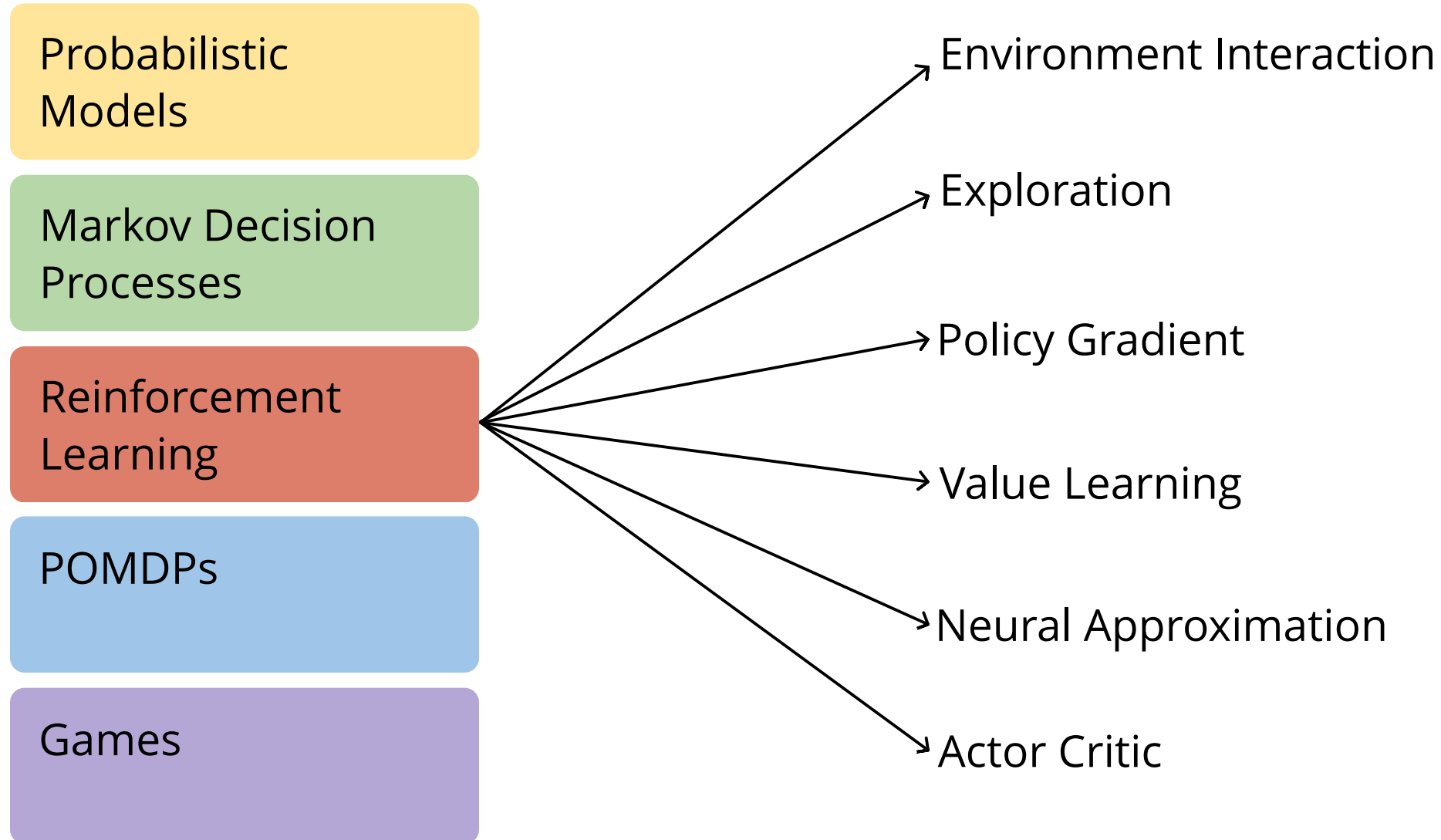


Sparse Sampling

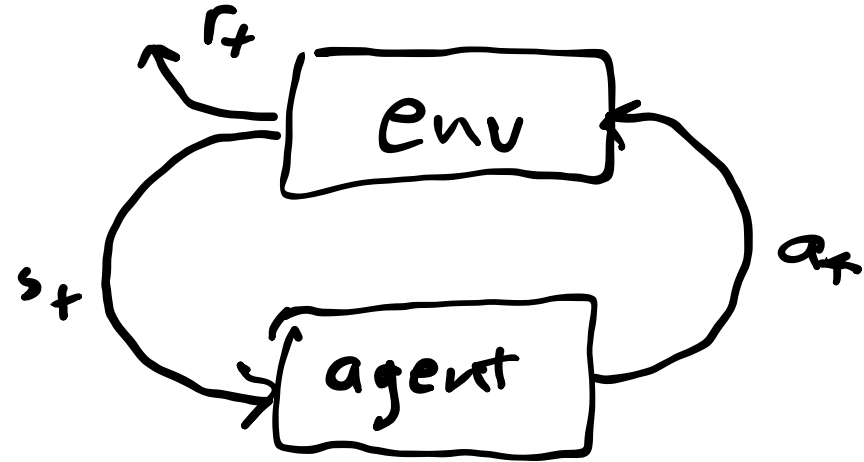


Guarantees *independent* of $|S|!!$

DMU Recap



Reinforcement Learning



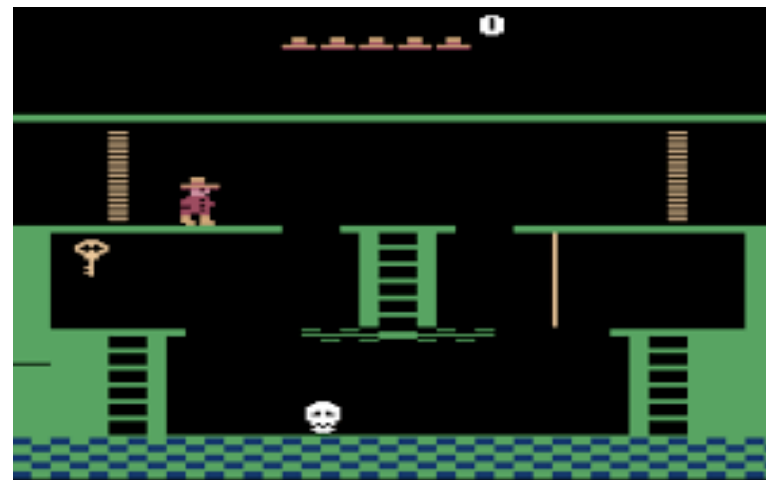
Challenges:

1. Exploration and Exploitation
2. Credit Assignment
3. Generalization

Exploration

Bandits

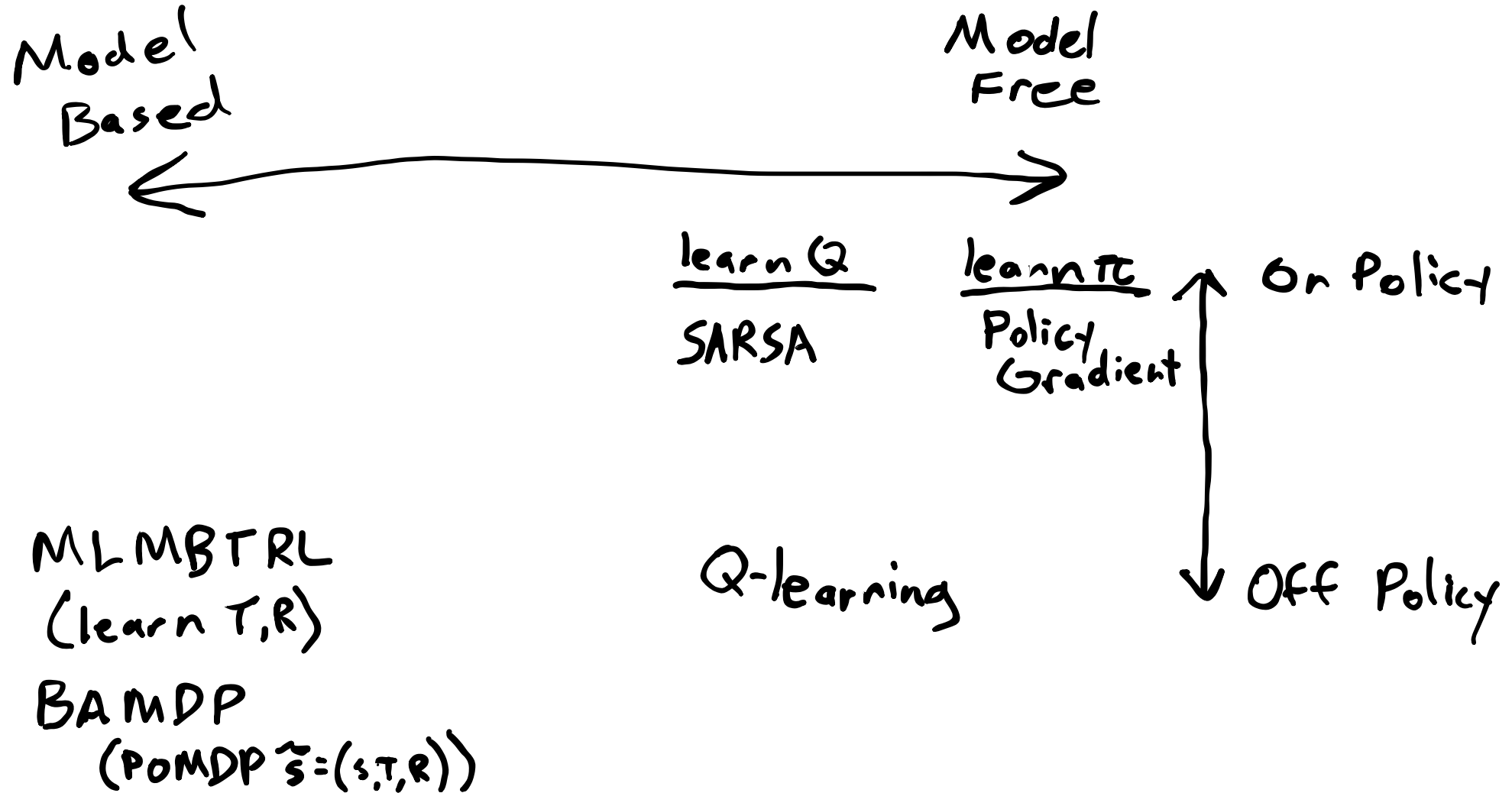
- ϵ -greedy
- softmax
- UCB
- Thompson Sampling
- Optimal DP Solution (solving a POMDP!)



Montezuma's Revenge!

- Pseudocounts
- Curiosity: extra reward for bad prediction
- Random network distillation

RL Algorithms

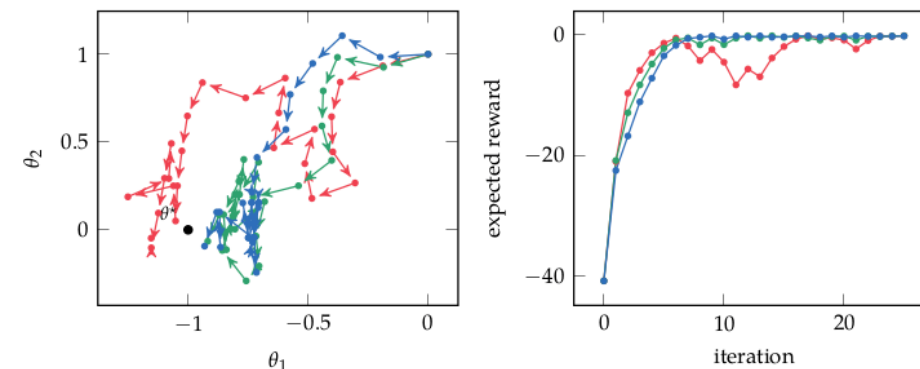


Policy Gradient

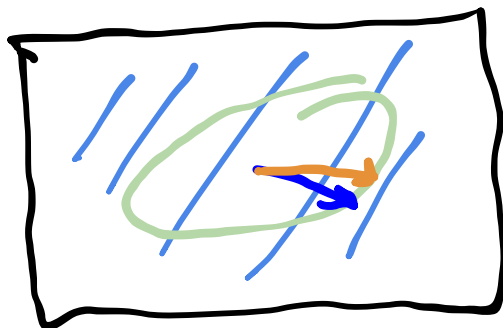
- Likelihood ratio trick
- Causality
- Baseline Subtraction

$$\nabla_{\theta} p_{\theta}(\tau) = p_{\theta}(\tau) \nabla_{\theta} \log p_{\theta}(\tau)$$

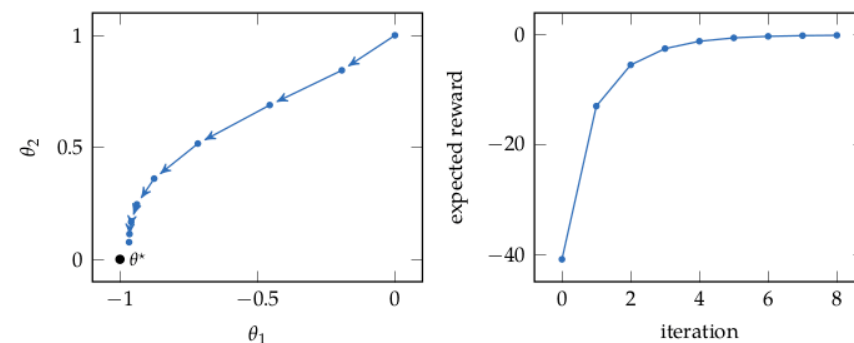
$$\nabla U(\theta) = \mathbb{E}_{\tau} \left[\sum_{k=1}^d \nabla_{\theta} \log \pi_{\theta}(a^{(k)} | s^{(k)}) \gamma^{k-1} (r_{\text{to-go}}^{(k)} - r_{\text{base}}(s^{(k)})) \right]$$



- Natural Gradient (TRPO)



KL div.
Bound



Q-Learning

SARSA

$$Q(s, a) \leftarrow Q(s, a) + \alpha(r_t + \gamma Q(s', a') - Q(s, a))$$

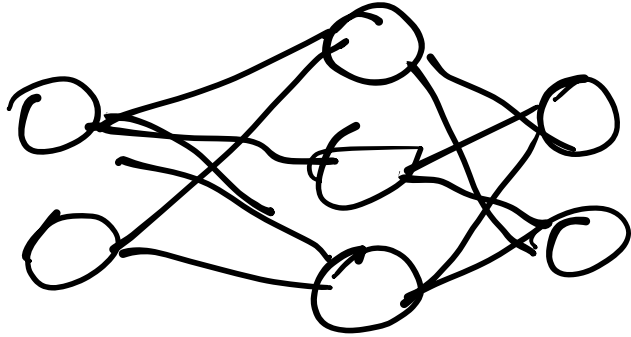
Eligibility Traces

Q-learning

$$Q(s, a) \leftarrow Q(s, a) + \alpha(r_t + \gamma \max_{a'} Q(s', a') - Q(s, a))$$

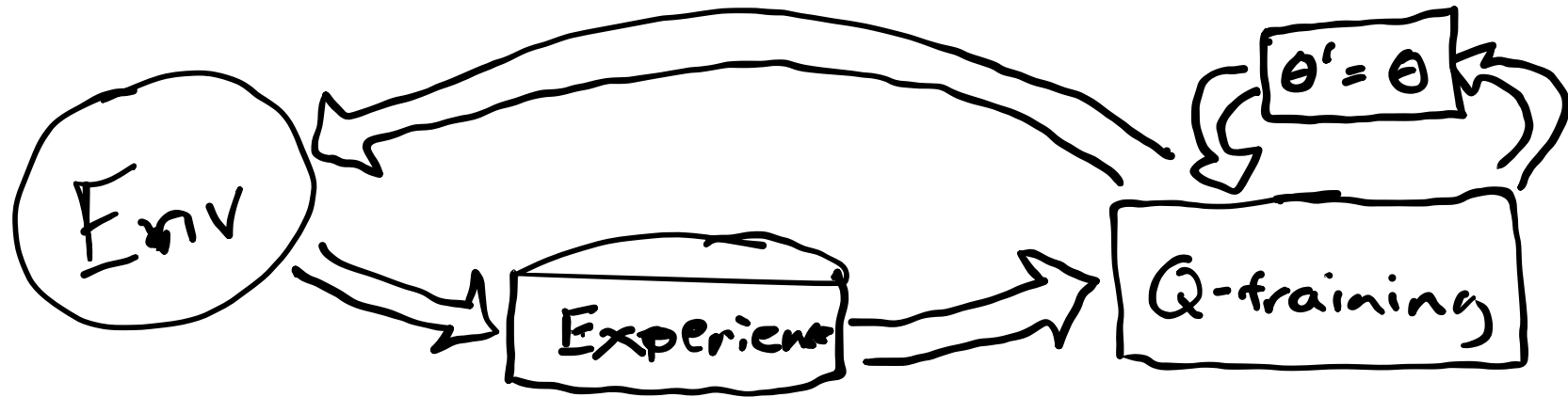
Double Q Learning

Neural Networks and DQN



$$f_{\theta}(x) = \sigma(W_2\sigma(W_1x + b_1) + b_2)$$

Backprop



Actor-Critic

- Actor: π_θ
- Critic: Q_ϕ, V_ϕ, A_ϕ

$$\widehat{\nabla U}(\theta) = \sum_{k=0}^d \nabla_\theta \log \pi_\theta(a_k | s_k) \gamma^k \hat{A}_k$$

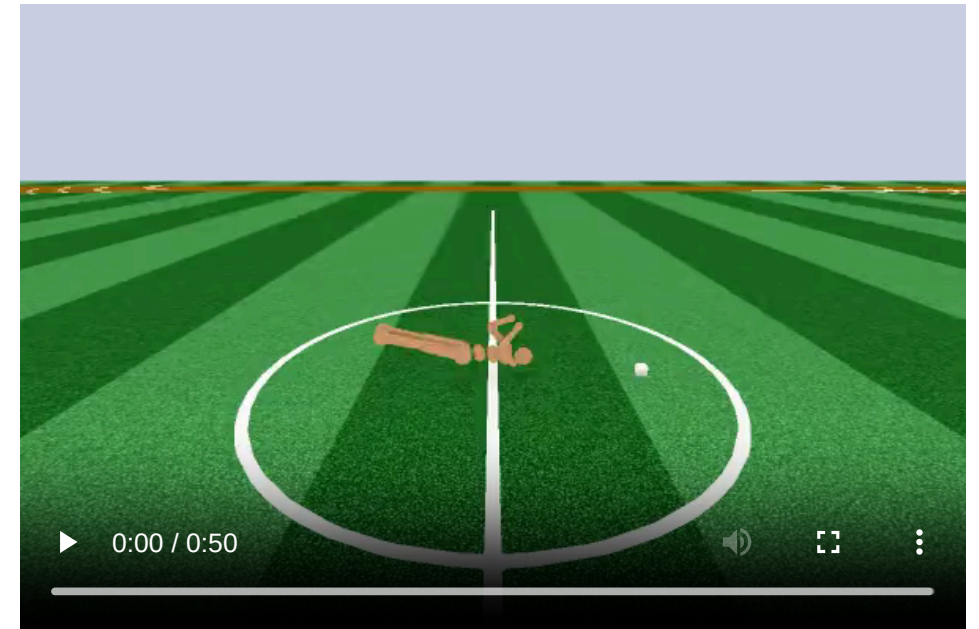
$$A_{\text{GAE}}(s_k, a_k) \approx \sum_{t=k}^{\infty} (\gamma \lambda)^{t-k} (r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t))$$

Soft Actor Critic (SAC)

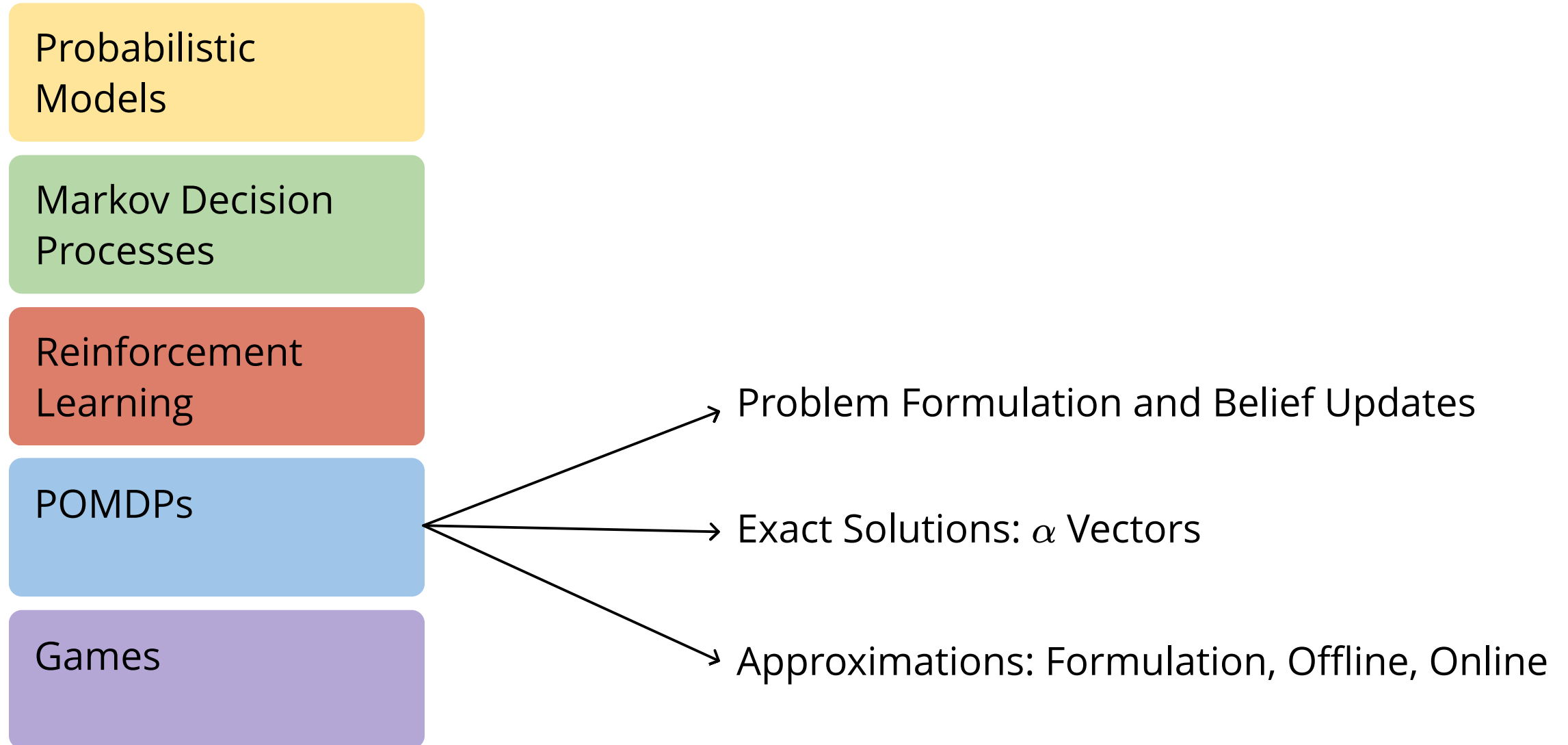
$$J(\pi) = E \left[\sum_{t=0}^{\infty} \gamma^t (r_t + \alpha \mathcal{H}(\pi(\cdot | s_t))) \right]$$

Proximal Policy Optimization (PPO)

$$E_{\substack{s \sim b_{\gamma, \theta} \\ a \sim \pi_\theta(\cdot | s)}} \left[\min \left(\frac{\pi_{\theta'}(a | s)}{\pi_\theta(a | s)} A_\phi(s, a), \text{clamp} \left(\frac{\pi_{\theta'}(a | s)}{\pi_\theta(a | s)}, 1 - \epsilon, 1 + \epsilon \right) A_\phi(s, a) \right) \right]$$

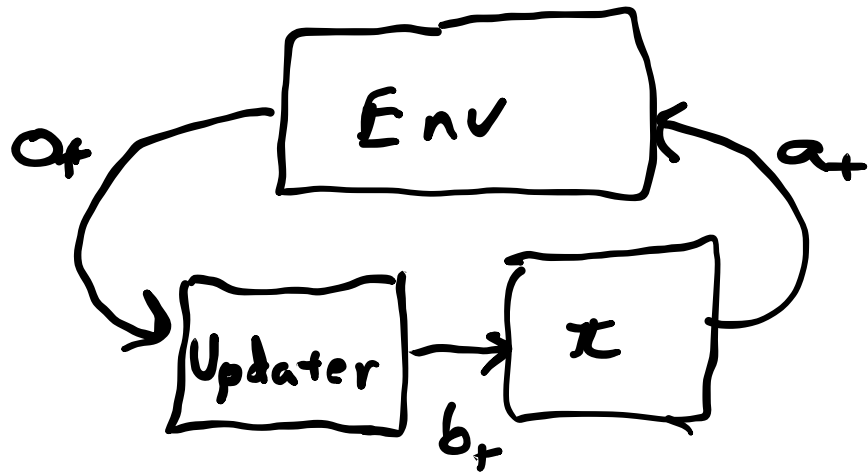


DMU Recap



POMDPs

$(S, A, T, R, O, Z, \gamma)$

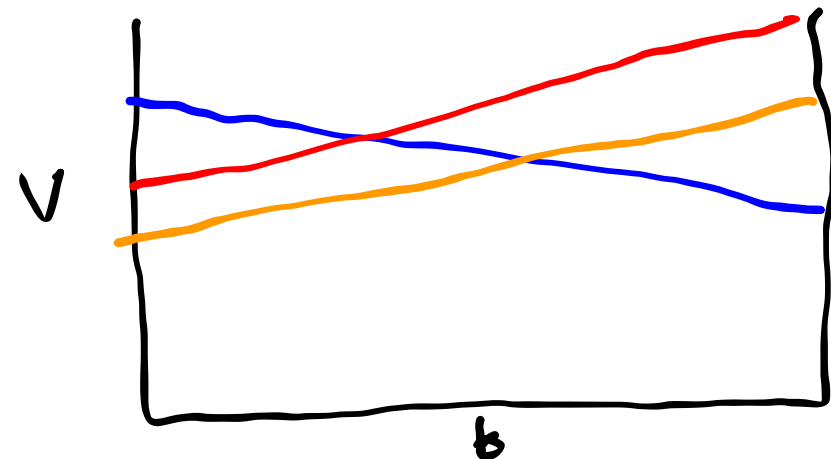


- Each alpha vector corresponds to a conditional plan
- You can prune alpha vectors by solving an LP

Belief Updates

- Discrete Bayesian Filter
- Particle Filter

Alpha Vectors



POMDP Approximations

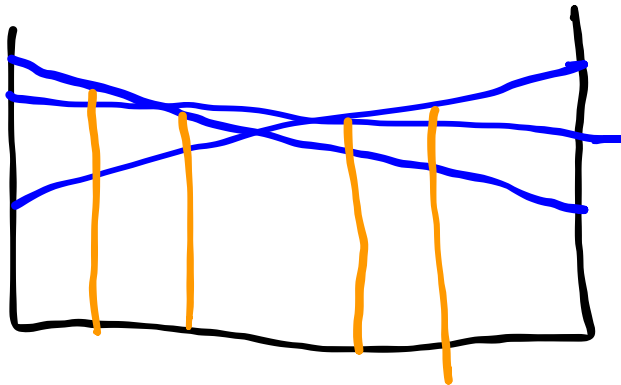
Formulation

- Certainty Equivalence
- QMDP
- Open Loop Feedback Control (OLFC)

Numerical

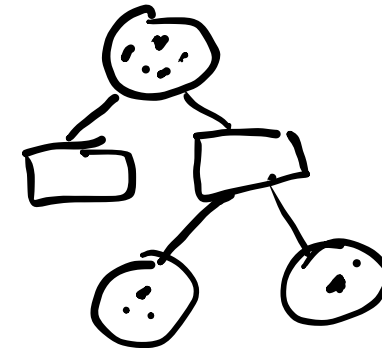
Offline

- Point-Based Value Iteration
- SARSOP



Online

- POMCP
- DESPOT



DMU Recap

Probabilistic
Models

Markov Decision
Processes

Reinforcement
Learning

POMDPs

Games

Normal Form Games and Nash Equilibria

Markov Games and POSGs

Normal Form Games

- ~~Optimal Solutions~~ No!
- Equilibria (e.g. Nash Equilibria)

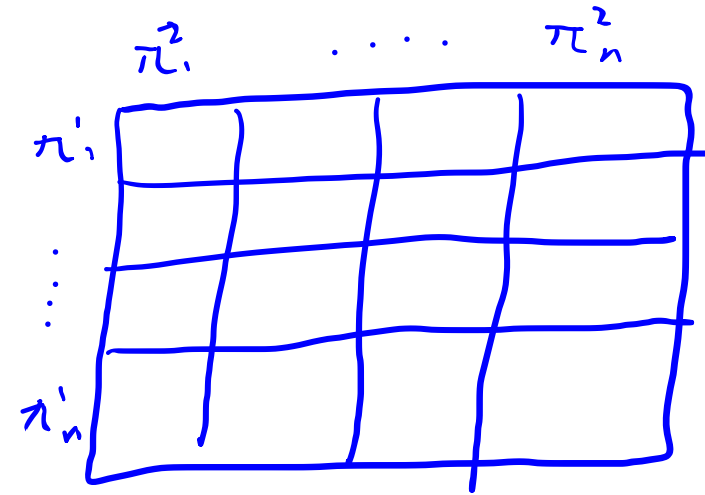
-1, -1	-3, 0
0, -3	-2, -2

- Every finite game has at least 1 Nash Equilibrium
- Might be pure or mixed
- Algorithms like fictitious play converge in special cases

Markov Games and POSGs

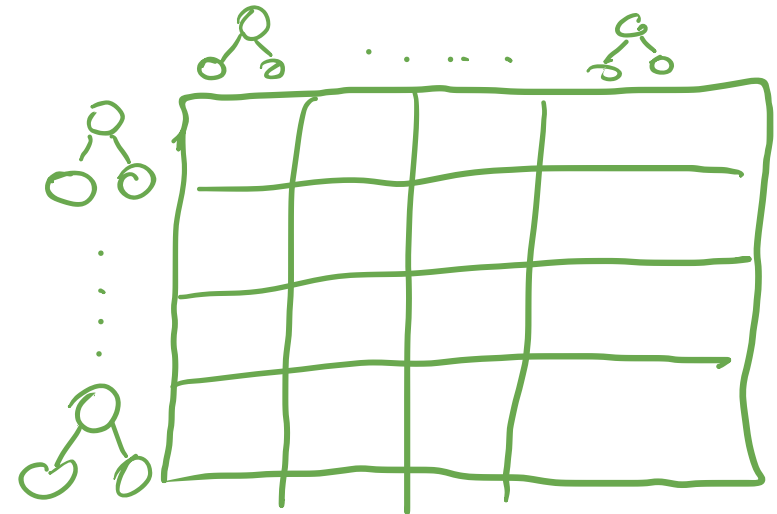
Markov Games (I, S, A, T, R, γ)

- All players play simultaneously
- Transitions are stochastic
- Best response involves solving an MDP
- Can be reduced to a simple game with policies as actions



Partially Observable Markov Games

- $(I, S, A, T, R, O, Z, \gamma)$
- Each player receives a noisy observation at each step
- Beliefs not practical to compute
- Can be reduced to simple game with policies as actions

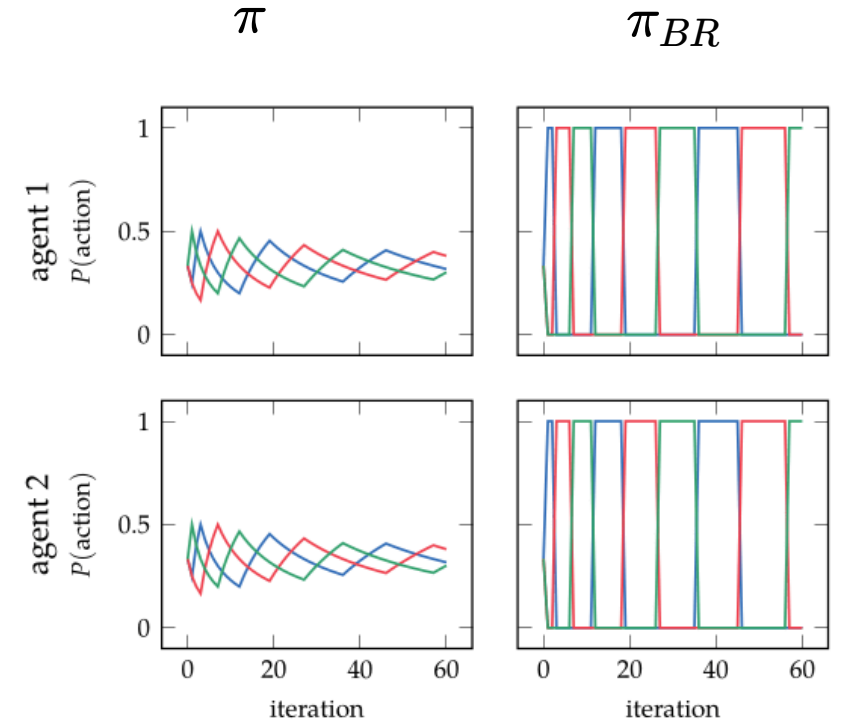


Fictitious Play in Markov Games

Loop:

1. $N(j, a^j, s) \leftarrow N(j, a^j, s) + \pi_{BR}^j(a^j \mid s) \quad \forall j, a^j, s$
2. $\pi^j(a^j \mid s) \propto N(j, a^j, s) \quad \forall j, a^j, s$
3. $\pi_{BR} \leftarrow$ best response to π

π (not necessarily π_{BR}) converges to a Nash equilibrium in many cases, notably 2-player zero-sum games



Open Questions after DMU

(Because we didn't cover them or no-one knows the answer!)

- How do we solve POSGs?
- How do we learn dynamics models to train on?
- Is imitation / supervised learning better than RL?
- How do we certify or build trust in these algorithms?
- Will AI replace or surpass human intelligence?

Recap

**After DMU you have basic tools to deal
with 4 Big Problems:**

1. Immediate and Future Rewards
2. Unknown Models
3. Partial Observability
4. Other Agents